STACKS, DISCRIMINANTS: LEMMA-DUALIZING-BASE-CHANGE-OF-FLAT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 589-593 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 595-646, the complete author proof (50 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 6 of queue_discriminant.json (pb=2323, stmt 589-593, proof 595-646).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00HN: sources/stacks-discriminant/pinned-original-source.tex, source lines 646–646. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00UJ: sources/stacks-discriminant/pinned-original-source.tex, source lines 646–646. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00HR: sources/stacks-discriminant/pinned-original-source.tex, source lines 646–646. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVB: sources/stacks-discriminant/pinned-original-source.tex, source lines 414–416. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BVC: sources/stacks-discriminant/pinned-original-source.tex, source lines 435–441. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BT3: sources/stacks-discriminant/pinned-original-source.tex, source lines 450–454. Cited by the retained statement or proof but not

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part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BT5: sources/stacks-discriminant/pinned-original-source.tex, source lines 493–497. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *If $A \to B$ is flat, then the base change map (0BVB) induces an isomorphism $\omega_{B/A} \otimes_B B_1 \to \omega_{B_1/A_1}$.*

Proof. If $A \to B$ is finite flat, then B is finite locally free as an A-module. In this case $\omega_{B/A} = \mathrm{Hom}_A(B, A)$ is the dual finite locally free A-module and formation of this module commutes with arbitrary base change which proves the lemma in this case. In the next paragraph we reduce the general (quasi-finite flat) case to the finite flat case just discussed.

Let $\mathfrak{q}_1 \subset B_1$ be a prime. We will show that the localization of the map at the prime $\mathfrak{q}_1$ is an isomorphism, which suffices by Algebra, Lemma 00HN. Let $\mathfrak{q} \subset B$ and $\mathfrak{p} \subset A$ be the prime ideals lying under $\mathfrak{q}_1$. By Algebra, Lemma 00UJ we can find an étale ring map $A \to A'$ and a prime ideal $\mathfrak{p}' \subset A'$ lying over $\mathfrak{p}$ such that $\kappa(\mathfrak{p}') = \kappa(\mathfrak{p})$ and such that

$$B' = B \otimes_A A' = C \times D$$

with $A' \to C$ finite and such that the unique prime $\mathfrak{q}'$ of $B \otimes_A A'$ lying over $\mathfrak{q}$ and $\mathfrak{p}'$ corresponds to a prime of C. Set $A'_1 = A' \otimes_A A_1$ and consider the base change maps (0BVB) for the ring maps $A \to A' \to A'_1$ and $A \to A_1 \to A'_1$ as in the diagram

$$\begin{array}{ccc} \omega_{B'/A'} \otimes_{B'} B'_1 & \longrightarrow & \omega_{B'_1/A'_1} \\ \uparrow & & \uparrow \\ \omega_{B/A} \otimes_B B'_1 & \longrightarrow & \omega_{B_1/A_1} \otimes_{B_1} B'_1 \end{array}$$

where $B' = B \otimes_A A'$, $B_1 = B \otimes_A A_1$, and $B'_1 = B \otimes_A (A' \otimes_A A_1)$. By Lemma 0BVC the diagram commutes. By Lemma 0BT3 the vertical arrows are isomorphisms. As $B_1 \to B'_1$ is étale and hence flat it suffices to prove the top horizontal arrow is an isomorphism after localizing at a prime $\mathfrak{q}'_1$ of B'_1 lying over $\mathfrak{q}$ (there is such a prime and use Algebra, Lemma 00HR). Thus we may assume that $B = C \times D$ with $A \to C$ finite and $\mathfrak{q}$ corresponding to a prime of C. In this case the dualizing module $\omega_{B/A}$ decomposes in a similar fashion (Lemma 0BT5) which reduces the question to the finite flat case $A \to C$ handled above. □